



Seat No.	
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Set	P
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S.E. (Civil) (Part – II) (Old-CGPA) Examination, 2018
ENGG. MATHEMATICS – III

Day and Date : Wednesday, 28-11-2018
Time : 2.30 p.m. to 5.30 p.m.

Max. Marks : 70

- N.B. :**
- 1) Figures to the **right** indicate **full** marks.
 - 2) **Use** of non-programmable calculator is **allowed**.
 - 3) Q. No. **1** is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. **3**. **Each** question carries **one** mark.
 - 4) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

(14×1=14)

1) The complete solution of $(D^2 - 2)^2 y = 0$ is

- a) $(C_1 + C_2 x)e^{\sqrt{2}x}$
- b) $C_1 e^{\sqrt{2}x} + C_2 e^{-\sqrt{2}x} + C_3 \cos \sqrt{2}x + C_4 \sin \sqrt{2}x$
- c) $(C_1 + C_2 x)e^{-\sqrt{2}x} + (C_3 + C_4 x)e^{\sqrt{2}x}$
- d) None of these

2) The particular integral of $(D^2 + 4)y = \cos 2x$ is

- a) $\frac{x \sin 2x}{4}$
- b) $\frac{x}{4} \cos 2x$
- c) $\frac{\sin 2x}{2}$
- d) $\frac{x \sin 2x}{2}$

3) The solution of $q = e^{-p}$ is

- a) $z = ax + by + c$
- b) $z = ax + e^{-a}y + c$
- c) $z = ax + e^a y + c$
- d) $z = ax + y^{-b} + c$

4) The solution of $(x^2 D^2 - xD)y = 0$ is

- a) $y = C_1 x + C_2 x^2$
- b) $y = C_1 + C_2 x^2$
- c) $y = C_1 + C_2 e^{2x}$
- d) $y = C_1 + C_2 (\log x)^2$

P.T.O.



- 5) Solution of $xp + yq = z$ is
a) $f(x^2, y^2) = 0$ b) $f(x, y) = 0$ c) $f(xz, yz) = 0$ d) $f\left(\frac{x}{y}, \frac{y}{z}\right) = 0$
- 6) If $L\{f(t)\} = \phi(s)$, then $L\{t f(t)\} =$
a) $\frac{d}{ds}[\phi(s)]$ b) $\frac{1}{s} \phi(s)$ c) $\int_s^\infty \phi(s) ds$ d) $-\frac{d}{ds}[\phi(s)]$
- 7) $L^{-1}\left\{\frac{1}{(s+3)^2}\right\} =$
a) $t e^{-3t}$ b) $t e^{3t}$ c) $t^2 e^{-3t}$ d) $t^2 e^{3t}$
- 8) The conditions for expansion of a function in a Fourier series are known as
a) Harmonic b) Riemann conditions
c) Periodic d) Dirichlet's conditions
- 9) The directional derivative of $\phi = xyz$ at $(1, 1, 1)$ is maximum in the direction of
a) $i - j + k$ b) $i + j + k$ c) $-i + j + k$ d) $i + j - k$
- 10) If $\vec{F} = (x + 3y) i + (y - 2z) j + (x + az) k$ is solenoidal, then $a =$
a) -2 b) 2 c) 1 d) 0
- 11) If mean of $x = 90$, the mean of $y = 50$, the coefficient of regression of x on y is 0.9 , then the equation of the line of regression of x on y is
a) $y = 0.9 x + 0.45$ b) $x = 0.9 y + 0.45$
c) $y = 0.9 x + 0.65$ d) $x = 0.9 y + 0.65$
- 12) The value of coefficient of correlation r lies between
a) -1 and 1 b) -2 and -1 c) 1 and 2 d) 2 and 3
- 13) A continuous random variable has following probability density function $f(x) = kx^2, 0 \leq x \leq 2$, then $k =$
a) $8/3$ b) $3/2$ c) $2/3$ d) $3/8$
- 14) For a binomial distribution with $n = 6$ and $p = \frac{1}{3}$, the mean and variance are
a) $2, 4$ b) $\frac{1}{3}, 3$ c) $2, \frac{4}{3}$ d) $2, \frac{3}{4}$



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**S.E. (Civil) (Part – II) (Old-CGPA) Examination, 2018
ENGG. MATHEMATICS – III**

Day and Date : Wednesday, 28-11-2018
Time : 2.30 p.m. to 5.30 p.m.

Marks : 56

- N.B. :**
- 1) Attempt **any three** questions from **each** Section.
 - 2) Figures to the **right** indicate **full** marks.
 - 3) **Use** of non-programmable is **allowed**

SECTION – I

2. a) Solve : $(D^3 - D^2 - 6D) y = x^2 + 1$. **3**
- b) Solve : $(D^2 - 4D + 1) y = e^{2x} \sin 2x$. **3**
- c) Solve : $(D^3 - D) y = 2 e^x + 2x + 1 - 4 \cos x$. **4**

OR

- c) Solve : $(D^2 + 4) y = \sec 2x$.
3. a) Solve : $x^2 \frac{d^3 y}{dx^3} + 3x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = x^2 \log x$ **4**
- b) The differential equation of a beam uniformly loaded with one end fixed and second subjected to a compressive force is given by **5**

$$EI \frac{d^2 y}{dx^2} + Py = -\frac{1}{2} Wx^2$$

where E, I, P and W are constants.

If $y = 0$ and $\frac{dy}{dx} = 0$ at $x = 0$, prove that $y = \frac{-W}{Pn^2} \cos nx - \frac{W}{2P} \left(x^2 - \frac{2}{n^2} \right)$

where $n^2 = P/EI$.

Set P



4. Attempt **any three** :

(3×3=9)

a) Find $L\{t e^{2t} \sin^2 t\}$

b) Find $L^{-1}\left\{\frac{s+2}{s(s+1)(s+3)}\right\}$

c) Evaluate $\int_0^{\infty} \frac{e^{-t} - e^{-3t}}{t} dt$ by using Laplace transform.

d) Find $L^{-1}\left\{\frac{1}{s(s^2+9)}\right\}$ by using convolution theorem.

5. a) Solve : $y^2 zp + x^2 zq = xy^2$.

3

b) Solve : $p^3 + q^3 = 27z$.

3

c) Solve : $3x \frac{\partial z}{\partial x} - 5y \frac{\partial z}{\partial y} = 0$ by the method of separation of variables.

3

SECTION – II

6. a) Find Fourier series to represent $f(x) = x^2$ in $(0, 2\pi)$ and hence deduce that

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

5

OR

a) Obtain Fourier expansion of $f(x) = e^x$ in $-\pi < x < \pi$.

5

b) Find cosine series of period 2π to represent $f(x) = \sin x$ in $0 \leq x \leq \pi$.

5

7. a) Find velocity and acceleration of a particle moving along the curve $x = 2\sin 3t$, $y = 2 \cos 3t$, $z = 8t$. Also find their magnitudes.

3

b) Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of $\overrightarrow{PQ}$ where Q is point $(5, 0, 4)$.

3

c) Show that the vector field $\vec{F} = 2xyz^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$ is irrotational. Find scalar potential ϕ such that $\vec{F} = \text{grad } \phi$.

3

Set P



8. a) The weight of 4000 students are found to be normally distributed with mean 50 kg and standard deviation 5 kg. Find the probability that a student selected at random will have weight (i) less than 45 kg (ii) between 45 and 60 kg.

[For S.N.V.Z, area under the curve between $Z = 0$ and $Z = 1$ is 0.3413 and that between $Z = 0$ and $Z = 2$ is 0.4772].

3

- b) Assume that the probability of an a coal miner being killed in a mine accident during a year is $\frac{1}{2400}$. Use Poisson's distribution to find the probability that in a mine employing 200 workers there will be at least one fatal accident in a year.

3

- c) Fit a binomial distribution to the following data :

3

Value of x :	0	1	2	3	4	5	6
Frequency f :	6	20	28	12	8	6	0

9. a) By the method of least squares, find the straight line that best fits the following data :

x :	1	2	3	4	5
f :	14	27	40	55	68

3

- b) Find the coefficient of correlation between x and y from the following data :
 $n = 20$, $\Sigma x = 80$, $\Sigma x^2 = 900$, $\Sigma y = 70$, $\Sigma y^2 = 800$, $\Sigma xy = 700$.

3

- c) Given : Mean of x = 50.07, Mean of y = 9.98

S. D. of x = 5.26, S. D. of y = 2.59

$r = 0.898$.

Find the equations of the lines of regression.

3

Set P

**SLR-TX – 45**

Seat No.	
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Set **Q**

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MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

(14×1=14)

- 1) The conditions for expansion of a function in a Fourier series are known as
 - a) Harmonic
 - b) Riemann conditions
 - c) Periodic
 - d) Dirichlet's conditions
- 2) The directional derivative of $\phi = xyz$ at (1,1, 1) is maximum in the direction of
 - a) $i - j + k$
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- 3) If $\vec{F} = (x + 3y) i + (y - 2z) j + (x + az) k$ is solenoidal, then $a =$
 - a) -2
 - b) 2
 - c) 1
 - d) 0
- 4) If mean of $x = 90$, the mean of $y = 50$, the coefficient of regression of x on y is 0.9 , then the equation of the line of regression of x on y is
 - a) $y = 0.9 x + 0.45$
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- 5) The value of coefficient of correlation r lies between
 - a) -1 and 1
 - b) -2 and -1
 - c) 1 and 2
 - d) 2 and 3

P.T.O.



- 6) A continuous random variable has following probability density function $f(x) = kx^2$, $0 \leq x \leq 2$, then $k =$
- a) $8/3$ b) $3/2$ c) $2/3$ d) $3/8$
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SECTION – I

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Set Q



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Set Q



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Set Q



SLR-TX – 45

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Set **R**

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MCQ/Objective Type Questions

Duration : 30 Minutes

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(14×1=14)1) Solution of $xp + yq = z$ is

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P.T.O.



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where E, I, P and W are constants.

If $y = 0$ and $\frac{dy}{dx} = 0$ at $x = 0$, prove that $y = \frac{-W}{Pn^2} \cos nx - \frac{W}{2P} \left(x^2 - \frac{2}{n^2} \right)$

where $n^2 = P/EI$.

Set R



4. Attempt **any three** :

(3×3=9)

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b) Find $L^{-1}\left\{\frac{s+2}{s(s+1)(s+3)}\right\}$

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b) Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of $\overline{PQ}$ where Q is point $(5, 0, 4)$.

3

c) Show that the vector field $\vec{F} = 2xyz^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$ is irrotational. Find scalar potential ϕ such that $\vec{F} = \text{grad } \phi$.

3

Set R



8. a) The weight of 4000 students are found to be normally distributed with mean 50 kg and standard deviation 5 kg. Find the probability that a student selected at random will have weight (i) less than 45 kg (ii) between 45 and 60 kg.

[For S.N.V.Z, area under the curve between $Z = 0$ and $Z = 1$ is 0.3413 and that between $Z = 0$ and $Z = 2$ is 0.4772].

3

- b) Assume that the probability of an a coal miner being killed in a mine accident during a year is $\frac{1}{2400}$. Use Poisson's distribution to find the probability that in a mine employing 200 workers there will be at least one fatal accident in a year.

3

- c) Fit a binomial distribution to the following data :

3

Value of x :	0	1	2	3	4	5	6
Frequency f :	6	20	28	12	8	6	0

9. a) By the method of least squares, find the straight line that best fits the following data :

x :	1	2	3	4	5
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- b) Find the coefficient of correlation between x and y from the following data :
 $n = 20$, $\Sigma x = 80$, $\Sigma x^2 = 900$, $\Sigma y = 70$, $\Sigma y^2 = 800$, $\Sigma xy = 700$.

3

- c) Given : Mean of x = 50.07, Mean of y = 9.98

S. D. of x = 5.26, S. D. of y = 2.59

$r = 0.898$.

Find the equations of the lines of regression.

3

Set R

**SLR-TX – 45**

Seat No.	
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Set	S
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S.E. (Civil) (Part – II) (Old-CGPA) Examination, 2018
ENGG. MATHEMATICS – III

Day and Date : Wednesday, 28-11-2018
Time : 2.30 p.m. to 5.30 p.m.

Max. Marks : 70

- N.B. :**
- 1) Figures to the **right** indicate **full** marks.
 - 2) **Use** of non-programmable calculator is **allowed**.
 - 3) Q. No. **1** is **compulsory**. It should be solved in **first 30 minutes** in Answer Book Page No. **3**. **Each** question carries **one** mark.
 - 4) **Answer MCQ/Objective type questions on Page No. 3 only. Don't forget to mention, Q.P. Set (P/Q/R/S) on Top of Page.**

MCQ/Objective Type Questions

Duration : 30 Minutes

Marks : 14

1. Choose the correct answer :

(14×1=14)

- 1) If $\vec{F} = (x + 3y) \mathbf{i} + (y - 2z) \mathbf{j} + (x + az) \mathbf{k}$ is solenoidal, then $a =$
a) -2 b) 2 c) 1 d) 0
- 2) If mean of $x = 90$, the mean of $y = 50$, the coefficient of regression of x on y is 0.9 , then the equation of the line of regression of x on y is
a) $y = 0.9x + 0.45$ b) $x = 0.9y + 0.45$
c) $y = 0.9x + 0.65$ d) $x = 0.9y + 0.65$
- 3) The value of coefficient of correlation r lies between
a) -1 and 1 b) -2 and -1 c) 1 and 2 d) 2 and 3
- 4) A continuous random variable has following probability density function $f(x) = kx^2$, $0 \leq x \leq 2$, then $k =$
a) $8/3$ b) $3/2$ c) $2/3$ d) $3/8$
- 5) For a binomial distribution with $n = 6$ and $p = \frac{1}{3}$, the mean and variance are
a) $2, 4$ b) $\frac{1}{3}, 3$ c) $2, \frac{4}{3}$ d) $2, \frac{3}{4}$

P.T.O.



- 6) The complete solution of $(D^2 - 2)^2 y = 0$ is
- $(C_1 + C_2 x)e^{\sqrt{2}x}$
 - $C_1 e^{\sqrt{2}x} + C_2 e^{-\sqrt{2}x} + C_3 \cos \sqrt{2}x + C_4 \sin \sqrt{2}x$
 - $(C_1 + C_2 x)e^{-\sqrt{2}x} + (C_3 + C_4 x)e^{\sqrt{2}x}$
 - None of these
- 7) The particular integral of $(D^2 + 4)y = \cos 2x$ is
- $\frac{x \sin 2x}{4}$
 - $\frac{x}{4} \cos 2x$
 - $\frac{\sin 2x}{2}$
 - $\frac{x \sin 2x}{2}$
- 8) The solution of $q = e^{-p}$ is
- $z = ax + by + c$
 - $z = ax + e^{-a}y + c$
 - $z = ax + e^a y + c$
 - $z = ax + y^{-b} + c$
- 9) The solution of $(x^2 D^2 - xD)y = 0$ is
- $y = C_1 x + C_2 x^2$
 - $y = C_1 + C_2 x^2$
 - $y = C_1 + C_2 e^{2x}$
 - $y = C_1 + C_2 (\log x)^2$
- 10) Solution of $xp + yq = z$ is
- $f(x^2, y^2) = 0$
 - $f(x, y) = 0$
 - $f(xz, yz) = 0$
 - $f\left(\frac{x}{y}, \frac{y}{z}\right) = 0$
- 11) If $L\{f(t)\} = \phi(s)$, then $L\{t f(t)\} =$
- $\frac{d}{ds} [\phi(s)]$
 - $\frac{1}{s} \phi(s)$
 - $\int_s^\infty \phi(s) ds$
 - $-\frac{d}{ds} [\phi(s)]$
- 12) $L^{-1}\left\{\frac{1}{(s+3)^2}\right\} =$
- $t e^{-3t}$
 - $t e^{3t}$
 - $t^2 e^{-3t}$
 - $t^2 e^{3t}$
- 13) The conditions for expansion of a function in a Fourier series are known as
- Harmonic
 - Riemann conditions
 - Periodic
 - Dirichlet's conditions
- 14) The directional derivative of $\phi = xyz$ at $(1, 1, 1)$ is maximum in the direction of
- $i - j + k$
 - $i + j + k$
 - $-i + j + k$
 - $i + j - k$



Seat No.	
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**S.E. (Civil) (Part – II) (Old-CGPA) Examination, 2018
ENGG. MATHEMATICS – III**

Day and Date : Wednesday, 28-11-2018
Time : 2.30 p.m. to 5.30 p.m.

Marks : 56

- N.B. :**
- 1) Attempt **any three** questions from **each** Section.
 - 2) Figures to the **right** indicate **full** marks.
 - 3) **Use** of non-programmable is **allowed**

SECTION – I

2. a) Solve : $(D^3 - D^2 - 6D) y = x^2 + 1$. **3**
- b) Solve : $(D^2 - 4D + 1) y = e^{2x} \sin 2x$. **3**
- c) Solve : $(D^3 - D) y = 2 e^x + 2x + 1 - 4 \cos x$. **4**

OR

- c) Solve : $(D^2 + 4) y = \sec 2x$.
3. a) Solve : $x^2 \frac{d^3 y}{dx^3} + 3x \frac{d^2 y}{dx^2} + \frac{dy}{dx} = x^2 \log x$ **4**
- b) The differential equation of a beam uniformly loaded with one end fixed and second subjected to a compressive force is given by **5**

$$EI \frac{d^2 y}{dx^2} + Py = -\frac{1}{2} Wx^2$$

where E, I, P and W are constants.

If $y = 0$ and $\frac{dy}{dx} = 0$ at $x = 0$, prove that $y = \frac{-W}{Pn^2} \cos nx - \frac{W}{2P} \left(x^2 - \frac{2}{n^2} \right)$

where $n^2 = P/EI$.

Set S



4. Attempt **any three** :

(3×3=9)

a) Find $L\{t e^{2t} \sin^2 t\}$

b) Find $L^{-1}\left\{\frac{s+2}{s(s+1)(s+3)}\right\}$

c) Evaluate $\int_0^{\infty} \frac{e^{-t} - e^{-3t}}{t} dt$ by using Laplace transform.

d) Find $L^{-1}\left\{\frac{1}{s(s^2+9)}\right\}$ by using convolution theorem.

5. a) Solve : $y^2 zp + x^2 zq = xy^2$.

3

b) Solve : $p^3 + q^3 = 27z$.

3

c) Solve : $3x \frac{\partial z}{\partial x} - 5y \frac{\partial z}{\partial y} = 0$ by the method of separation of variables.

3

SECTION – II

6. a) Find Fourier series to represent $f(x) = x^2$ in $(0, 2\pi)$ and hence deduce that

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

5

OR

a) Obtain Fourier expansion of $f(x) = e^x$ in $-\pi < x < \pi$.

5

b) Find cosine series of period 2π to represent $f(x) = \sin x$ in $0 \leq x \leq \pi$.

5

7. a) Find velocity and acceleration of a particle moving along the curve $x = 2\sin 3t$, $y = 2 \cos 3t$, $z = 8t$. Also find their magnitudes.

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Set S



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